# Heuristics and trade-offs

## Optimization · Level 4 · Lesson 4



# In this lesson you will

- Explain NP-hardness in plain words and why heuristics are used.
- Implement 2-opt and compare metaheuristics for escaping local optima.
- Measure the optimality gap and put fairness limits in the open.

# Key ideas

- Build fast, then improve with local search
- Escape local optima: annealing, tabu, genetic
- Gap = (found - bound) ÷ found
- Several goals: show the trade-off with numbers

# Activity: Route race

1. Build (6 min): Nearest neighbour on a paper map.
2. Improve (8 min): Uncross with 2-opt moves.
3. Check (6 min): Compare with the best route.
4. Fair (5 min): Add a limit on the longest ride.

**Use a real local map if you can; travel times work as well as distances. Work in pairs and swap routes to check each other’s sums.**

# Let’s discuss

1. Why do companies accept routes nobody has proved optimal?
2. Which metaheuristic would you try first for a timetable, and why?
3. Who should decide the longest acceptable ride on a school bus?

# Check yourself

- Your route is 52 km and a bound says 50 km. Write the gap, and one sentence a manager would understand.